Colin D Lewis

DEMAND FORECASTING and INVENTORY CONTROL

A computer aided learning approach

THE INSTITUTE OF OPERATIONS MANAGEMENT

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Demand forecasting and inventory control

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OPERATIONS
MANAGEMENT

WOODHEAD PUBLISHING LIMITED

Published by Woodhead Publishing Ltd in association with the Institute of Operations Management

Woodhead Publishing Ltd
Abington Hall, Abington
Cambridge CB1 6AH, England

First published 1997, Woodhead Publishing Ltd

British Library Cataloguing in Publication Data
A catalogue record for this book is available from the British Library.

ISBN 1 85573 241 6

Designed by Geoff Green (text) and The ColourStudio (cover)
Typeset by BookEns Ltd, Royston, Herts.
Printed by T J International Ltd, Padstow, Cornwall, England.

To Maven and Emma and in memory of Mark

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Preface

This book allows both students and practitioners to understand the theory and practice of current demand forecasting methods, the links between forecasts produced as a result of analysing demand data and the various methods by which this information, together with stocked item cost information, is used to establish the controlling parameters of the most commonly used inventory control systems.

The demand forecasting section of the book concentrates on the family of short-term forecasting models based on the exponentially weighted average and its many variants and also a group of medium-term forecasting models based on a time series, curve fitting approach. Although the forecasting models described are chosen primarily for their proven ability to forecast future demand, they can be used to analyse any variable whose values occur at regular intervals of time.

The inventory control sections of the book investigate the re-order level policy (or fixed order quantity system) and re-order cycle policy (or periodic review system) and indicate how these two policies, provided with the appropriate forecasting and cost information, can be operated at minimum cost while offering a high level of customer service. Consideration is also given to the hybrid (s,S) inventory control policy and to the situation characterised by items such as slow moving spares, where the demand for a single unit can be expressed by an average interval between issues as long as twenty years.

Which of the many demand forecasting models and inventory control policies available is most appropriate for particular stocked items is discussed in the context of a Pareto analysis approach whereby such items are allocated into 'A', 'B' and 'C' categories based on annual usage value.

The accompanying integrated, computer aided learning (CAL) package OPSCON (OPerationS planning and CONtrol) allows readers to create demand data files or use prepared data files for subsequent forecasting analysis. Following this forecasting analysis, a unique feature of the package allows the user, on terminating the forecasting analysis within the forecasting package FOREMAN, to 'carry' the resulting forecast information as demand input to the inventory control package STOCKMAN's two major inventory control policies. These two policies are represented by Monte Carlo simulations of what would have occurred in practice over a twelve year period assuming an imposed monthly demand and cost environment. As a result of the simulation analysis, recommendations are made for those values of the parameters controlling the chosen inventory policy which would offer a high level of service to customers at a minimum cost. Since all the necessary calculations are performed automatically, for the practitioner in particular, the OPSCON package when provided with demand and cost information can be used to establish the correct inventory control parameters without the user necessarily having acquired any knowledge of the underlying theory.

The integrated CAL package OPSCON occupies about 1.2 Mbytes of disk space and can be installed on and run either from the hard disk of any IBM-compatible PC within a DOS or WINDOWS environment or on a local area network (LAN). The package is totally menu driven with each menu composed of options with unique first capitalised (upper case) characters making each choice available with a single key depression. This allows the package to be used just as easily by the uninitiated as those who have studied the accompanying text. The installation and operating instructions for OPSCON for those who can't wait to get started are in Appendix A!

COLIN LEWIS

Part I

Forecasting

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1

Demand forecasting and inventory control

Introduction

Forecasting is a necessary pre-requisite to most operational activities. Without an estimate of the future it is not possible to plan for the level of activity which is to be expected and, hence, not possible to estimate the resources that need to be designed, planned and controlled to fulfil that level of activity.

Inventory control is the science-based art of controlling the amount of inventory (or stock) held, in various forms, within an organisation to meet the demand placed upon that business economically. To control the amount of inventory, it is clearly necessary to forecast the level of future demand, where such demand can be regarded as essentially either independent or dependent.

Independent demand describes the type of demand for a product or service which is independent of demand for other apparently related products or services. Many services organisations, public utilities and retail organisations display such independent demand characteristics. Within a supermarket, for instance, it is unlikely that the demand for bread is dependent on the demand for ice cream. Also, even within manufacturing environments – where demand at lower levels of the production process are clearly dependent on demand at higher levels – from a stock holding point-of-view it may be more sensible to ignore this self-evident dependency and assume that trends identified in past data – using a forecasting procedure – could be a more reliable (and certainly a more cost effective) indicator of future demand (i.e. that it is assumed that demand is independent). This apparent conundrum will be explained later in this chapter.

However, generally within manufacturing organisations, where a planned level of production of finished products is the norm, the demand for sub-assemblies, components and raw materials that make up the finished product is clearly highly dependent on that planned level of production. Hence in automobile manufacture, the demand for wheels, tyres, etc, is essentially a multiple of five times the demand for completed vehicles (i.e. four road wheels plus a spare). The methods of controlling the provisioning of sub-assemblies, components and raw materials for such dependent demand situations are radically different from those for independent demand.

Within a typical dependent demand environment, given a commitment to a future production plan of a specified number of finished products at specified times (Master Production Schedule) and also given that information is available such that the number of sub-assemblies, components and raw materials that make up each finished product can be defined (Bill of Material), combined with information as to the current stock holding situation of these items (Status of all Materials), the future demand for sub-assemblies, components and raw materials in terms of their required quantities and timing can theoretically be forecast with a high degree of certainty and accuracy. The process which converts a production plan of finished products into a schedule of supply of sub-assemblies, components and raw materials is referred to by the generic term Material Requirements Planning (MRP) which is essentially a large, and often very complicated, database manipulation exercise which requires large amounts of information regarding the relationships between finished products and their constituent sub-assemblies, components and raw materials. Given this information, which also needs continually updating if any engineering changes are introduced, demand at all levels of the production process can theoretically be established by exploding the planned demand of finished products at the highest level into sub-assemblies, components and raw materials at progressively lower levels and aggregating the results over all levels of the production process, while also taking into account existing stocks. Although simple in concept, the successful application of MRP to a production schedule of a mix of finished products within an existing inventory situation and subject to the vagaries of the production processes involved and the in-built inaccuracies of the information held in the Bill of Material is a target for which many manufacturing companies aim, but few achieve completely.

Given that such a dependent demand process could establish future demand at all levels of production with complete accuracy, in theory there would no need for any stocks of sub-assemblies, components or raw materials since their delivery could be synchronised to coincide exactly with their requirement. Hence the adoption of terms such as Just in Time (JIT) and Zero Inventory. The reality however is that in practice such an ideal situation is virtually impossible to achieve, although the best practitioners claim to come close.

Within a dependent demand environment, at the lower levels in such a production process, rather than relying on an expensive and complicated database exercise which involves the process of exploding and aggregating demand over several levels of production, if past demand can be assumed to be a reasonable indicator of future demand (i.e. demand can be assumed to be independent) stocks held by traditional inventory control policies with parameters established through a forecasting regime may well be cost effective. Certainly no automotive manufacturer establishes the demand for the many standard nuts, bolts, washers and fasteners through an MRP process – even though technically it should be possible. Even for such relatively cheap items, however, it is still necessary for the inventory control policies which are used to control such items do so effectively with a high level of service but at a reasonable cost.

This book and its accompanying CAL package OPSCON are concerned solely with the control of items whose demand characteristics are independent. Readers whose area of interest covers dependent demand situations are recommended to consult Harrison's[2] *Finite capacity scheduling – the art of synchronized manufacturing* and Kenworthy's[5] *Planning and control of manufacturing processes.*

Forecasting versus prediction

In general terms, forecasting is interpreted as being a scientific process of estimating a future event by casting forward past data. The past data are initially analysed to establish the underlying trends which characterise the data and this information is then used in a predetermined way to obtain an estimate of the future. Prediction, however, is generally interpreted as a process of estimating a future event based on subjective considerations. However, a scientifically produced forecast, established on the assumption that characteristic trends identified in past data will continue into the future, should always be open to alteration in the light of changes in market conditions. Examples of such changes could be:

- it is predicted, in advance of the event, that such trends are unlikely to be continued due to legislative changes which are assumed to affect future demand, or
- after the event, some extraneous causal effect is detected which invalidates the assumption of continuity of demand.

However, because predictions are predominantly subjective, they are generally more expensive to produce on a routine basis than forecasts. Hence, where many product lines are involved, it is generally more effective to operate on the assumption that scientifically produced forecasts are assumed to be satisfactory unless a monitoring procedure indicates that the forecast for a particular product line or item is no longer in control, at this time predictions of future demand may well be required to re-establish the validity of the forecast.

Different types of forecasting methods

One way of classifying or categorising forecasting methods is to define the type of forecast on the basis of the time period associated with the demand data which are being analysed, as illustrated in Table 1.1.

Short-term forecasting

Although there is no strict demarcation between the various forms of forecasting categorised within Table 1.1, it is generally assumed that short-term forecasting will often be associated with many product lines or items, as typically occurs in an inventory control environment. Within an inventory environment, it is also true that the demand patterns being analysed are relatively fast moving with an average per period in excess of twenty such that the normal probability distribution can be assumed to represent the distribution of demand per unit time. The forecasting models used when operating in such an environment must be simple and relatively cheap to operate while still being robust. The family of forecasting models based on the exponentially weighted average, originally suggested by Holt,[3] has proved to meet these criteria more satisfactorily

Table 1.1 Categorisation of type of forecast based on the underlying time unit of data involved

Category of type of forecast	Time period associated with the data being analysed	Example of forecasting application	Forecasting techniques used
Immediate-term	$\frac{1}{4}$ hr to 1 day	Electricity demand forecasting	Various
Short-term	1 week to 1 month	Demand forecasting in industry and commerce	Exponentially weighted averages and derivatives
Medium-term	1 month to 1 year	Sales and financial forecasting	Regression, curve fitting, time series analysis
Long-term	1 year to 1 decade	Technological forecasting	DELPHI, think tanks, etc

than any other group of models and has traditionally become the basis for forecasting in many inventory control situations because of its:

- computational cheapness in terms of processing time and storage requirements;
- robustness and ability for the forecasting process to be monitored so that 'out of control' situations can be detected;
- ease of 'starting up' when including new items with no previous demand history;
- adaptability in terms of changing sensitivity in line with the characteristic of the demand data encountered; and
- flexibility in terms of ability to cope with stationary, growth and seasonal demand patterns.

Medium-term forecasting

Increasingly as computer power becomes cheaper more sophisticated forecasting models, which require more complicated calculations than the exponentially weighted average family of forecasting models, are becoming financially viable as a method of forecasting for many stocked items. Forecasting methods such as:

- curve fitting and regression;
- Fourier analysis; and
- Baysian forecasting;

now feature in some of the more advanced, professionally developed software for inventory control.

Forecast horizon, fitting and forecasting

The forecast horizon is the number of time periods ahead of the known data over which forecasts are calculated (i.e. extrapolation). In many situations the maximum forecast horizon is about six periods ahead, since the confidence with which forecasts any further ahead of this can be made is likely to be low. An exception to this general rule is when a strong seasonal influence is known to exist, in which case forecasts up to a whole season ahead might well be justified i.e. up to twelve months for monthly data.

Fitting is the process of producing a forecasting model which fits known data (i.e. interpolation) whereas forecasting is the process of extrapolating a fitted model into the future, i.e. ahead of known data.

Within an environment where forecasting models are based on parameters which can be adjusted, a good fit is usually established by adjusting the value of those parameters to minimise the Sum of Squared forecasting Errors (SSE) or the Mean Squared forecasting Error (MSE).

It is generally assumed that the best fitted model will also be the best model for forecasting where, in its strict interpretation, forecasting produces future estimates ahead of known data.

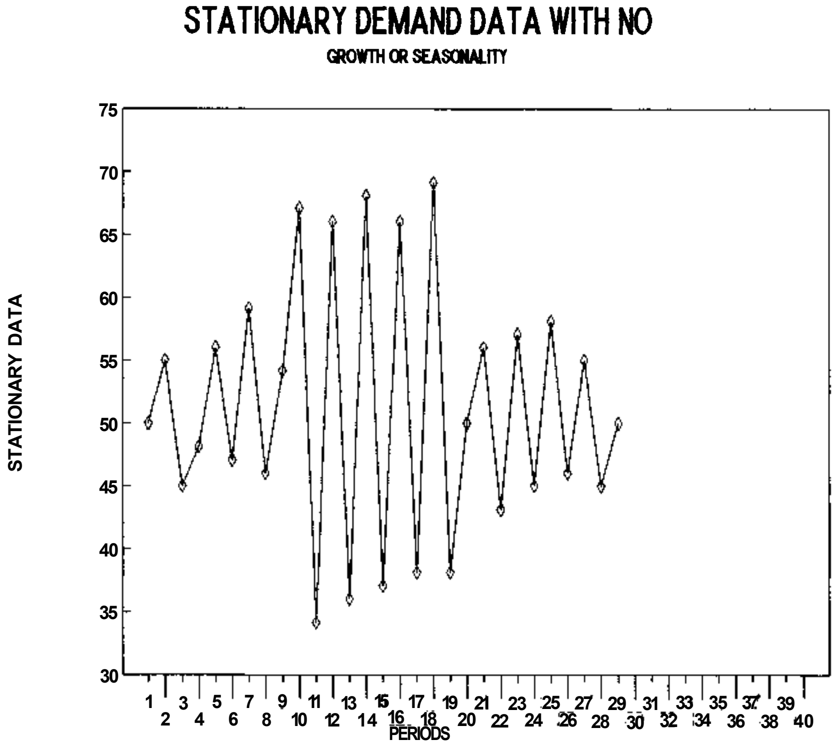
Characteristics of customer demand patterns requiring forecasting

Inventory control systems must cope with a variety of different customer demand patterns (for which forecasts are necessary) if an effective overall policy for controlling inventory (or stocks) is to be achieved. In practice, in ascending order of complexity, it is assumed that the following demand patterns can exist:

1. **Stationary demand** – assumes that although customer demand per unit time fluctuates, there is no apparent long-term underlying growth or seasonal trend. Figure 1.1 illustrates the basic stationary character of such data but also identifies the fact that variability in demand exists. Because no growth or seasonality is assumed in stationary demand patterns, forecasts ahead are fixed in value and the forecast for one period ahead is the forecast for any number of periods ahead.

   Although within a stationary demand pattern no growth or seasonality is assumed to exist, it should be accepted that occasionally fundamental changes in the demand pattern may occur, but these are presumed to be short-term in nature, such as:
   - impulses – individual demands which are significantly higher or lower than normal. Such impulses are best ignored by a forecasting system linked to an inventory control policy, since such policies are basically designed to cope with a reasonably stable level of demand with a known, measurable degree of variation; and
   - step changes – a series of successive demands which are significantly higher or lower than normal. The ideal response of a forecast to a step change in demand is that it should react as quickly as possible. Should this not be feasible, a competent forecasting system should at least identify that such a step change has occurred and should also instigate remedial action to ensure that the forecast, which will naturally lag behind such a sudden change of level, is corrected.

   Figure 1.2 illustrates a demand pattern where a single period

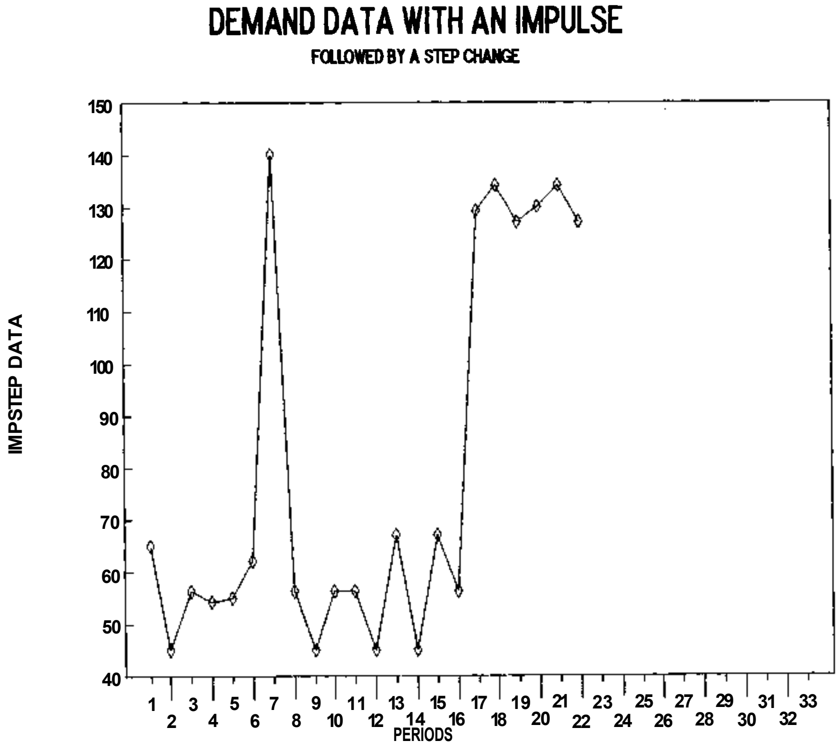


1.1 Stationary demand pattern with no long-term growth or seasonality but with considerable variability of demand. This data series is available within OPSCON as STATIONARY.

impulse (a significant, high demand occurring for one period only) is followed by a positive step in demand (a succession of significantly high values).

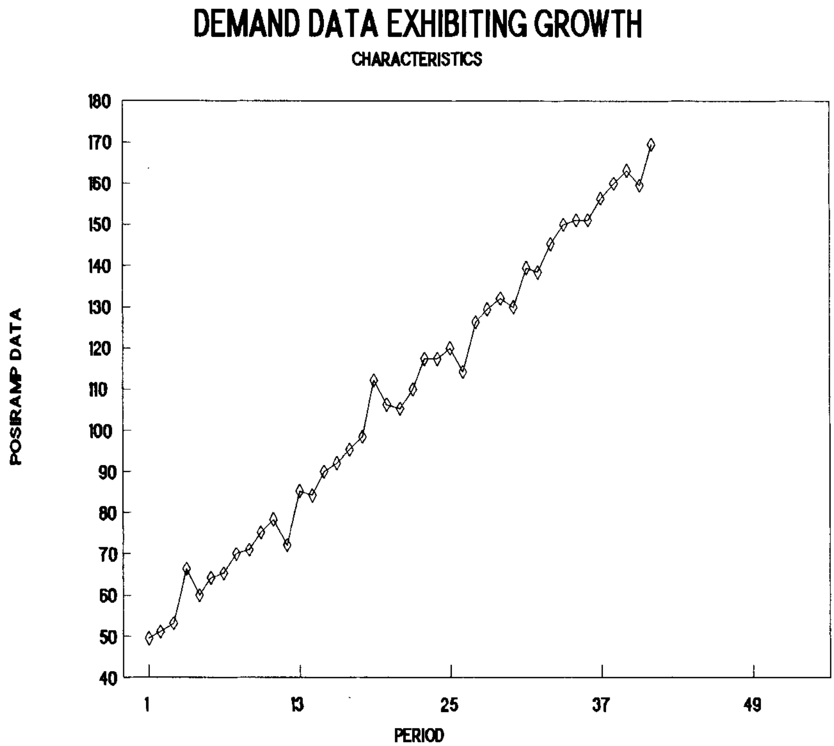
2. **Demand with growth characteristics** – where a demand pattern exhibits a growth characteristic over the longer term, the forecasting models required to accommodate that growth become more complex than those used in the stationary demand situation discussed earlier. Such inappropriate stationary forecasting models produce fitted forecasts which lag behind the data together with forecasts ahead which are fixed in value. Figure 1.3 illustrates a demand pattern where demand is growing long term.

NOTE: It is recognised that demand may decline rather than grow, in which case this continual drop in demand can be regarded as negative growth.



1.2 A stationary demand pattern where an impulse is followed by a change step. This data series is available within OPSCON as IMPSTEP.

3. **Demand with seasonal characteristics** – many demand series are influenced by the seasons of the year and by other events which occur annually. In such situations it is possible to establish the degree to which demand in any particular period of the year (i.e. month, quarter or accounting period) is higher or lower than for a typical average period. Hence the aim of forecasting models taking seasonality into account is to establish this relationship for each and every period within the year and to use the de-seasonalising factors that are identified by this process to produce forecasts. For technical reasons it is generally assumed that growth may also exist in demand patterns characterised by seasonality as is shown in Fig. 1.4. If there is no growth, the analysis simply registers actual growth as negligible.



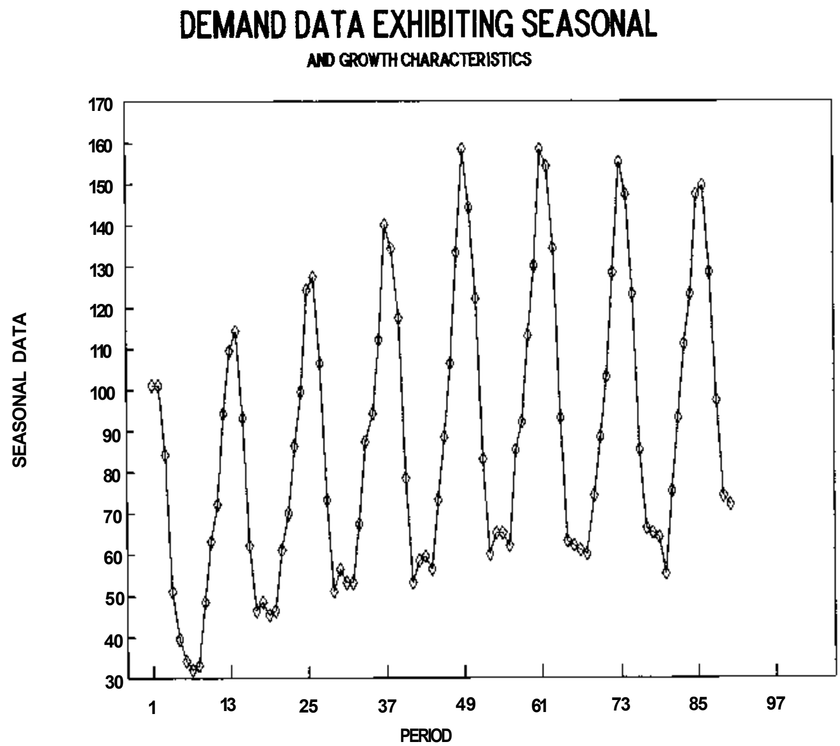
1.3 A demand pattern characterised by long-term growth. This data series is available within OPSCON as POSIRAMP.

Inventory control

The benefits of holding stock

To be viable, the holding of inventory (or stocks) must offer benefits to the customers whose demand requirements are met by the stock holding process. Some of the principal benefits of holding stock are:

1. The demands of customers are buffered by the stock holding process from the suppliers of replenishment orders to the stock holding process. Thus, given that the holding of stock is organised economically, the frequent small demand orders of customers can be met more effectively by the supply of large, infrequent replenishment orders by suppliers. Thus stock can often act as an effective cushion against interruptions to supply. Although not immediately thought of in terms of an item



1.4 A demand pattern characterised by long-term growth and seasonality. This data series is available within OPSCON as SEASONAL.

controlled by orthodox inventory control procedures, stocks of drinking water facilitated by the construction of reservoirs clearly assist in the process of meeting customers' demand throughout the year, even though the supply of water through natural rainfall is effectively temporarily interrupted during the summer months.

2. Because inventory systems are replenished with relatively large orders, the unit costs of the items provisioned by such large orders can often be reduced due to:

- longer production runs absorbing set-up costs more effectively;
- discounts or price breaks offered for large orders; and
- more economical transport and packaging costs.

Inventory operating costs

To offset the benefits of holding stocks, it must always be recognised that the stock holding process incurs specific operating costs, the more important of which are:

1. Ordering costs – which include the costs involved in raising replenishment orders which may involve the paperwork involved together with the associated labour costs. These can be reduced drastically through computerisation and the use of EDI (Electronic Data Interchange) but are never negligible.
2. Storage (or holding) costs – which are generally expressed as a percentage of the unit cost of the item involved (typically 20–30% pa) and made up of:

   - the costs of borrowing money to invest in stock or alternatively the loss of possible interest that could have been made had a company not used its own money to invest in stock (typically 10–15% pa);
   - the costs directly associated with storing goods, i.e. storemen's wages, rates, heating and lighting, store's transport, racking and palletisation, protective clothing, weighing equipment, etc;
   - costs involved in preventing deterioration of stock; (typically 0–30% pa);
   - costs incurred as a result of obsolescence which could include costs of re-work or scrapping (typically 4–7% pa); and
   - costs of insurance to cover fire, theft and third party injury (typically about 1.5% pa).

3. Stockout costs – costs which are incurred as a result of not being able to satisfy customer demand when stocks are not available. These costs are generally very difficult to define in terms of whether they should be assessed on a basis of the actual number of units of shortage or the time period over which the shortage occurs. In view of this difficulty, most inventory control models ignore stockout costs and argue that they are provided for, if somewhat obliquely, by offering a reasonable level of service to customers of the inventory system.
4. Administration costs – which can cover a multitude of fixed costs which are only incurred because stocks are held.

Inventory control policies

To run a successful inventory system, which should effectively balance the supply of replenishment orders against the requirements of demand, it is clearly necessary to have a policy which defines a series of rules as to how and when replenishment orders are to be raised. Essentially the raising of replenishment orders can be triggered either when stock-on-hand (the physical stock held plus any outstanding replenishment orders less committed stock) reaches a certain level or alternatively replenishment orders can be raised on a regular time basis i.e. once a month, once a quarter, etc.

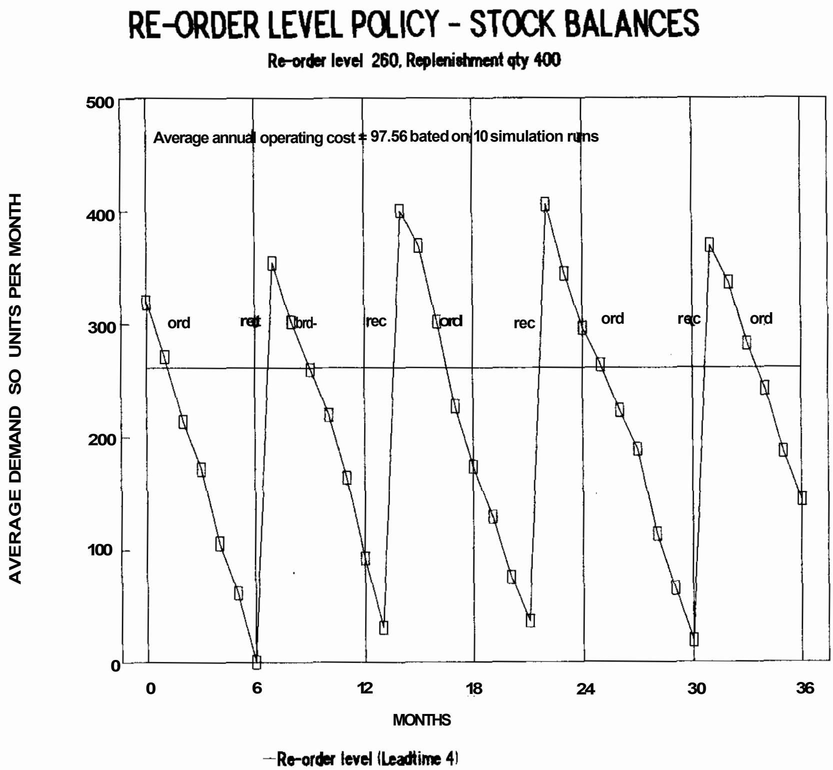
If replenishment orders are raised on a level basis, to maintain reasonably balanced stocks, the size of the replenishment order is usually fixed in size leading to a policy known as the re-order level (or re-order point) policy within which both the value of the re-order level has to be specified (see Chapter 7) together with a specification of the size of the replenishment quantity (see Chapter 8).

Alternatively, if replenishment orders are placed on a regular time basis, because the level of stocks at the time the replenishment order is placed will vary considerably, it is usual to place variable size replenishment orders; larger orders being placed when stock levels at review are low and conversely smaller replenishment orders when stock levels at review are high. The simplest and most effective method of arranging such a replenishment ordering policy is to raise an order whose size is established on the basis of a specified maximum stock level less the level of stock held at the time of review. Such a policy is referred to as a re-order cycle policy (see Chapter 9).

Both the re-order level and re-order cycle policies have advantages and disadvantages and these have to be considered in detail when deciding which type of policy is most appropriate for a particular stocked item. However, whichever policy is chosen, it is essential that the parameters controlling the operation of the policy (re-order level and replenishment quantity in the case of the re-order level policy and maximum stock level and review period in the case of the re-order cycle policy) are set at the correct levels such that the policy offers a reasonable level of service in terms of meeting demand without doing so at excessive cost.

Such control parameters are clearly going to be dependent on the characteristics of the demand imposed on the stock control system and the delay in replenishment orders being delivered (leadtime). This is demonstrated clearly by graphs of stock balances generated by the OPSCON package as shown in Fig. 1.5 to 1.7.

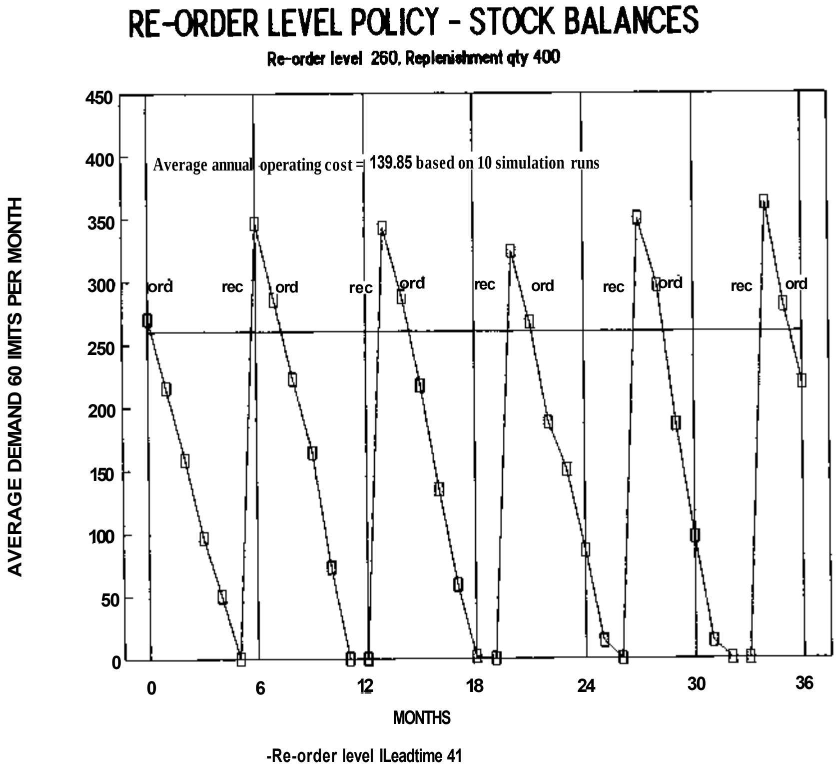
In Fig. 1.5 it can be seen that, when controlling stock with a re-order level policy, the stock balances displayed indicate that the policy appears to be operating reasonably effectively in that enough stock is held so that stockouts (periods when demand cannot be satisfied) are likely to be relatively few while on the other hand stocks never become excessively large. In this situation the average demand per month has been established



1.5 A correctly tuned stock control policy showing stock balances satisfying demand at minimum cost.

at 50 units. For a particular stocked item, and operating the policy with a re-order level of 260 and a replenishment quantity of 400 subject to a delivery leadtime of 4 months this produces an annual average total operating cost of 97.56. Clearly the parameters controlling this inventory policy have been well chosen in the light of the demand and replenishment leadtimes imposed on the system, and the total annual operating costs (storage, ordering and stockout costs) can be assumed to be as low as is possible in this demand environment.

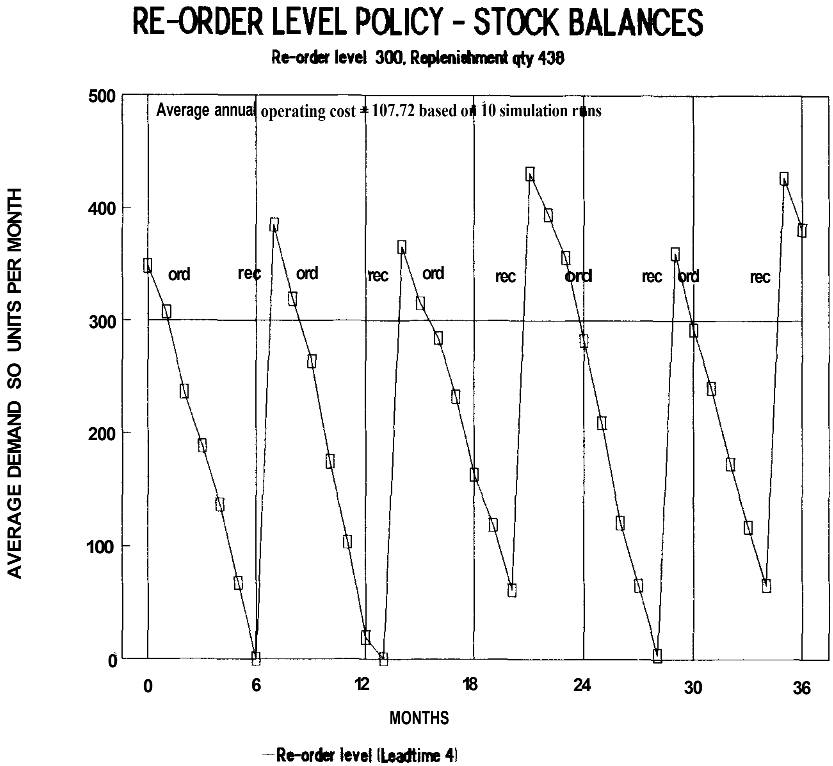
Figure 1.6 shows the stock balances for the same re-order level inventory policy operating under the same controlling parameter values as shown in Fig. 1.5, but now in a situation where the average demand per unit time has increased by 20%. Here, the increase in demand has led to extremely low stocks, which has inevitably created frequent stockouts and hence a significant increase in stockout (or penalty) costs. These increased penalty costs have not been compensated for by a relatively small drop in



1.6 An incorrectly tuned stock control policy where a 20% increase in demand has not been accommodated by a resetting of the controlling parameters. This has led to a shortage of stock and resulting stockouts.

storage costs, resulting in an overall increase in the annual average total operating cost from 97.56 to 139.85.

To regain a minimum cost operation in this new situation, it would be necessary to alter the values of the parameters controlling the inventory policy (say by increasing the re-order level and the replenishment quantity) in the light of the increased demand. Figure 1.7 demonstrates this situation, where to compensate for the increased demand the re-order level has been raised from 260 to 300 units and the replenishment quantity from 400 to 438 units. The resulting average annual operating cost in this situation is 107.72, significantly lower that the 139.85 incurred in the previous situation.



1.7 Stock balances for a stock control policy where the controlling parameters have been correctly adjusted to compensate for a 20% increase in average demand.

Conclusion

In more general terms, it should now be apparent that the parameters controlling inventory polices are:

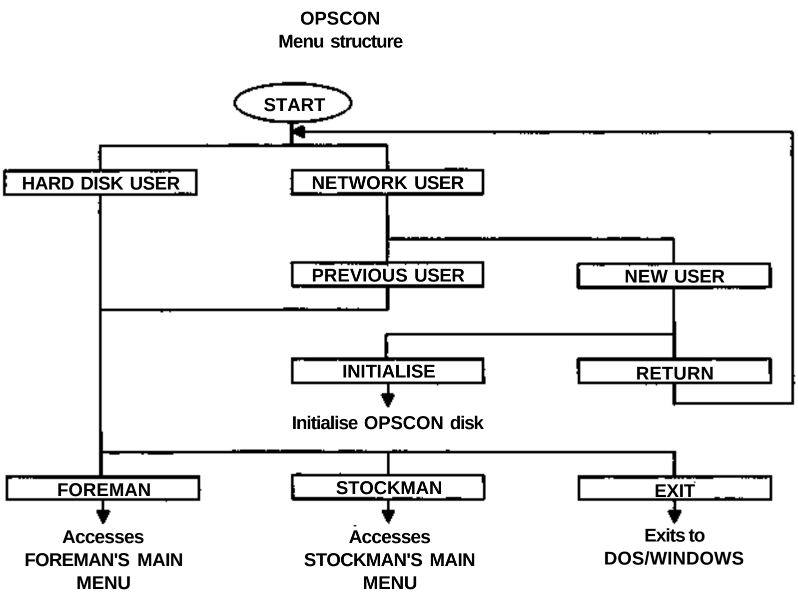
- for the re-order level policy, the re-order level value and the size of the replenishment order; and
- for the re-order cycle policy, the review period and the maximum stock level used to evaluate replenishment quantity values.

The setting of these parameters must be linked to the level of demand imposed on the inventory system and also to the delay between placing replenishment orders and their subsequent receipt (i.e. the leadtime).

This relationship between forecasting the level of demand and using such forecasts together with estimates of the replenishment leadtime to adjust (or fine tune) the controlling parameters of inventory policies is the essence of inventory control theory and thus of this book and its associated CAL package.

The OPSCON CAL module

The OPSCON CAL module contains the two inter-related packages FOREMAN and STOCKMAN. The OPSCON module can be run on either a standard PC by a HARD-DISK USER or on a network by a NETWORK USER.

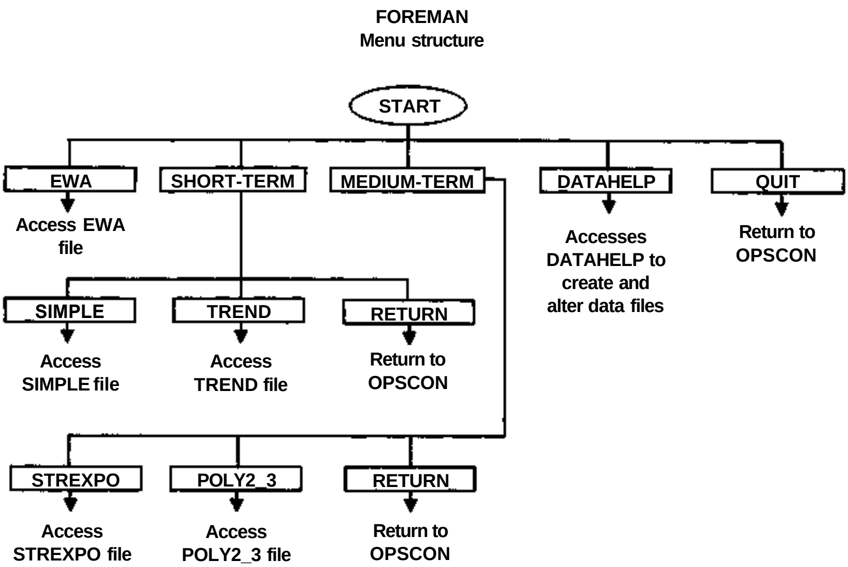


For network users it is assumed that all temporary files to be saved by the OPSCON module, and also graphical .PIC files specified by the user, will be saved to a floppy disk located in the A:\ drive. The disk used for this purpose needs to be initialised and is subsequently referred to as the OPSCON disk.

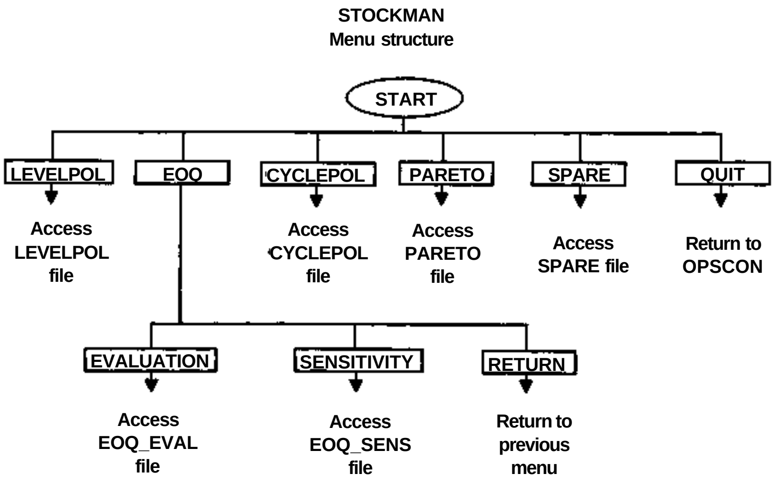
Packages within OPSCON associated with this chapter

FOREMAN package – contains six files for analysing demand series and producing forecasts as follows:

- EWA – a file detailing the calculations involved in the simple exponentially weighted average, adaptive and delayed adaptive response rate forecasting models;



- SIMPLE – a file containing three short-term forecasting models based on the simple exponentially weighted average, adaptive and delayed adaptive response rate methods. Any demand series can be analysed and three sets of DEMONSTRATION data are provided;
- TREND – a file containing three short-term forecasting models based on Brown's double smoothed exponentially weighted average and Holt Winters' quarterly and monthly seasonal methods. Any demand series can be analysed and three sets of DEMONSTRATION data are provided;
- STREXPO – a file containing two medium-term forecasting models based on a straight line and exponential growth trend curves. Quarterly and monthly seasonal analysis can be initiated once a trend has been chosen. Any demand series can be analysed and three sets of DEMONSTRATION data are provided;
- POLY2_3 – a file containing two medium-term forecasting models based on second and third order polynomial trend curves. Quarterly and monthly seasonal analysis can be initiated once a trend has been chosen. Any demand series can be analysed and three sets of DEMONSTRATION data are provided; and
- DATAHELP – a file providing facilities for creating and altering data files suitable for analysis by the forecasting files SIMPLE, TREND, STREXPO and POLY2_3.



STOCKMAN package – contains six files for simulating inventory control policies and demonstrating relationships within inventory control, as follows:

- LEVELPOL – a file which simulates the operation of a re-order level inventory policy. Options allow the user to change policy parameters, view resulting graphs and alter costs;
- EOQ_EVAL – a file which progressively develops the formulation of the Economic Order Quantity. Once the formulation is complete cost and demand variables can be altered to study the effect on the EOQ;
- EOQ_SENS – a file which compares numerically and graphically the costs involved in operating at a replenishment order quantity other than the EOQ;
- CYCLEPOL – a file which simulates the operation of a re-order cycle inventory policy. Options allow the user to change policy parameters, view resulting graphs and alter costs;
- PARETO – a file consisting of 250 stocked items which can be sorted either alphabetically by part number or in descending order of value to demonstrate the Pareto or ABC relationship; and
- SPARE – a file demonstrating the costs involved for an inventory control situation of expensive, slow moving spares where the demand may be as low as only once every twenty years.